AWS VPC PEERING PROJECT

EWB Cloud Labs — Project Report

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AWS Region: Europe (Stockholm) — eu-north-1

Evidence date: 30 October 2026

Project objective

Configure two Amazon VPCs, connect them using VPC peering, route traffic between private subnets, and validate access using a bastion host and security-group-controlled ICMP tests.

Outcome

The supplied evidence shows an active VPC peering connection, three running EC2 instances, successful ping tests to App-A and App-B, SSH access through the bastion to a private instance, and a controlled ICMP failure followed by successful recovery.

Prepared from the screenshots supplied for this project. Resource configuration and test outcomes are reported as shown in that evidence.

1. Architecture and resource inventory

The lab uses two VPCs in the same AWS Region. VPC A contains the public bastion subnet and a private App-A subnet. VPC B contains the private App-B subnet. The active peering connection provides a private route between the VPC CIDR ranges.

Resource	Configuration / observed value
VPC A	ewb-vpc-a — 10.10.0.0/16
VPC B	ewb-vpc-b — 10.20.0.0/16
Public subnet A	ewb-a-public — 10.10.1.0/24
Private subnet A	ewb-a-private — 10.10.2.0/24
Private subnet B	ewb-b-private — 10.20.1.0/24
Bastion	ewb-bastion — private IP 10.10.1.10; public IP shown in evidence: 16.170.229.38
App-A	ewb-app-a — private IP 10.10.2.10; no public IPv4 shown
App-B	ewb-app-b — private IP 10.20.1.10; no public IPv4 shown
Peering connection	ewb-a-b-peer — pcx-0370124e0cefd7ad8 — Active

2. Connectivity and security validation

The bastion is used as the entry point. App-A and App-B remain private and are reached using their private IPv4 addresses. Security-group rules restrict SSH and ICMP to specified source addresses rather than opening SSH to all IPv4 addresses.

Observed test results

Test	Evidence / result
Bastion → App-A (10.10.2.10)	4 packets received; 0% packet loss.
Bastion → App-B (10.20.1.10)	4 packets received; 0% packet loss.
SSH through bastion to private App-B	Terminal prompt identifies ip-10-20-1-10.
Controlled ICMP failure	App-A ping to App-B showed 100% packet loss after the relevant ICMP rule was removed.
ICMP recovery	After the rule was restored, App-A received 4 replies from App-B; 0% packet loss.

Note: A separate pre-peering baseline-failure screenshot was not included in the supplied images, so this report does not claim that specific screenshot as evidence.

3. Screenshot evidence

The following figures reproduce the supplied AWS Console and terminal screenshots. Captions describe only what is visible in the evidence.

Figure 1. VPC peering connection

The peering connection named ewb-a-b-peer is shown as Active.

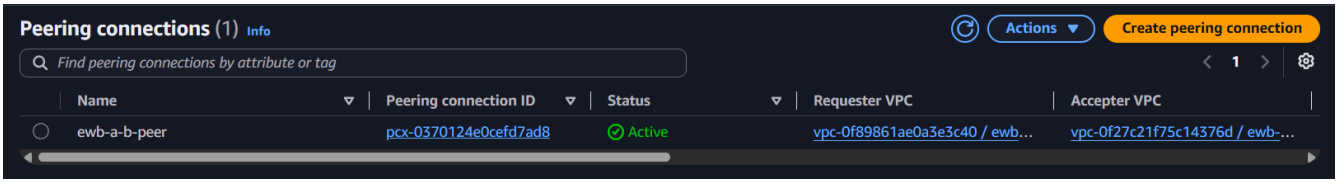


Figure 2. EC2 instance inventory

The bastion, App-A, and App-B instances are running and show successful status checks.

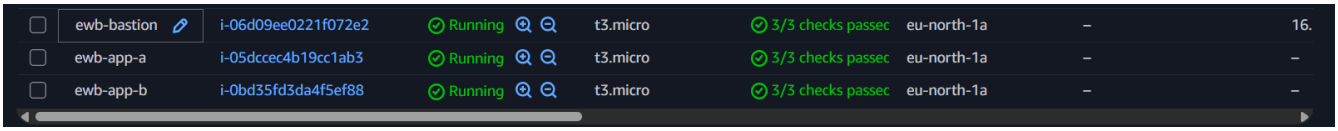


Figure 3. Bastion network details

The bastion has private IPv4 address 10.10.1.10 and public IPv4 address 16.170.229.38 in this capture.

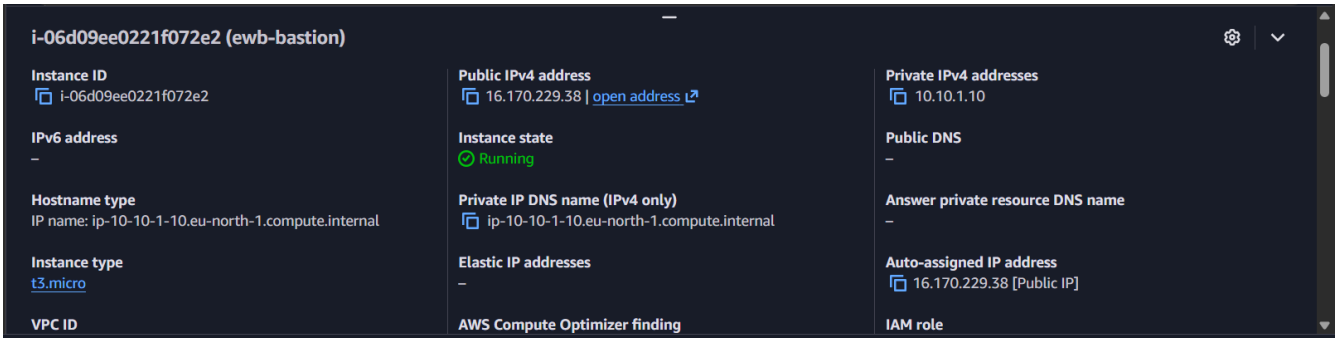


Figure 4. App-A network details

App-A has private IPv4 address 10.10.2.10 and no public IPv4 address.

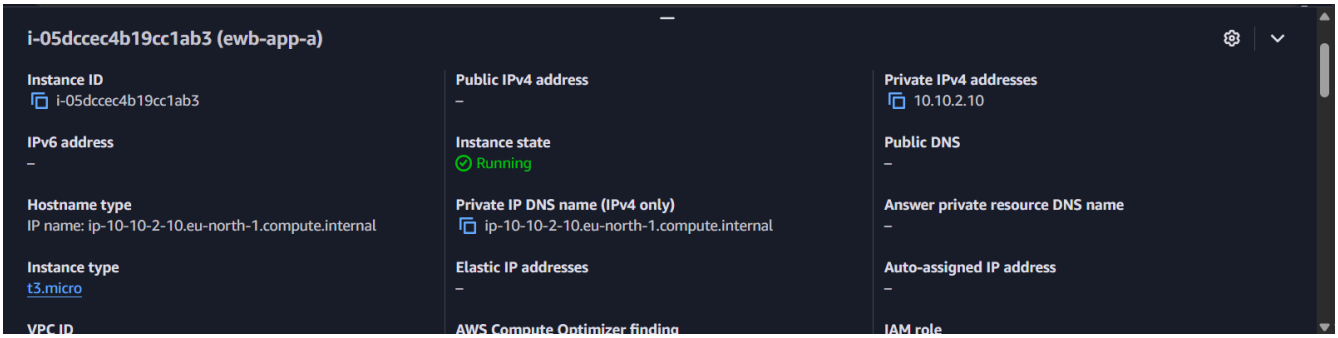


Figure 5. App-B network details

App-B has private IPv4 address 10.20.1.10 and no public IPv4 address.

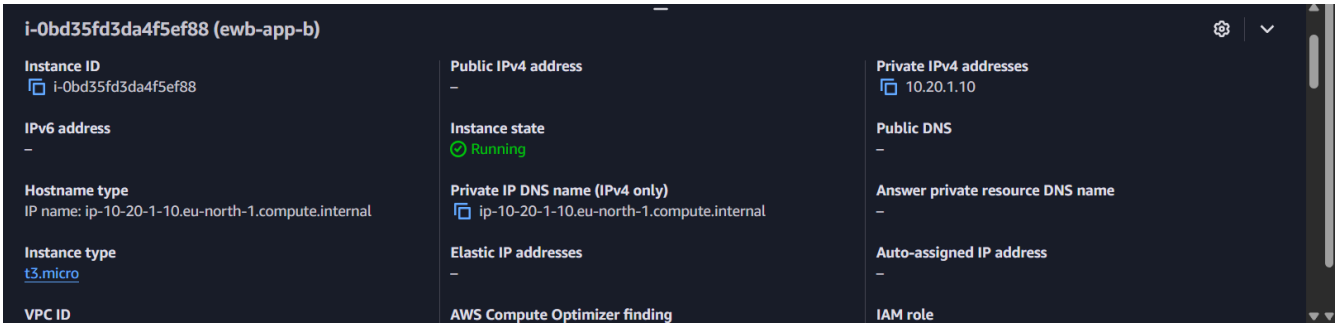


Figure 6. Private route table for VPC A

The screenshot identifies ewb-rt-a-private and its subnet association.



Figure 7. Private route table for VPC B

The screenshot identifies ewb-rt-b-private and its subnet association.



Figure 8. Bastion security group

SSH access is restricted to the recorded client public IPv4 address with a /32 source.

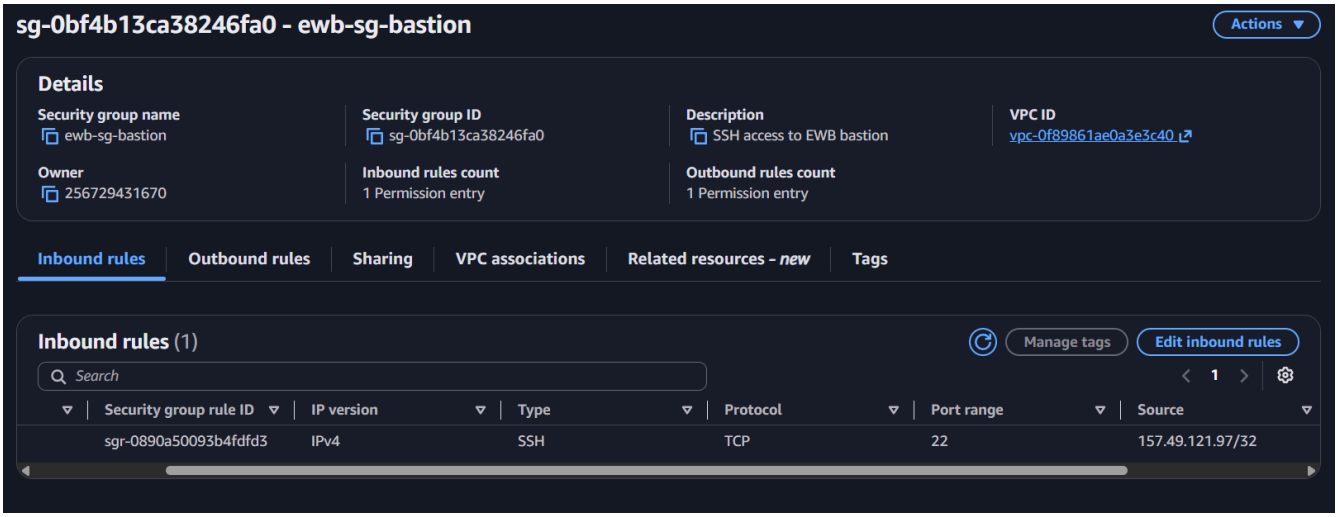


Figure 9. App-A security group

Rules show SSH and ICMP sources for App-A.

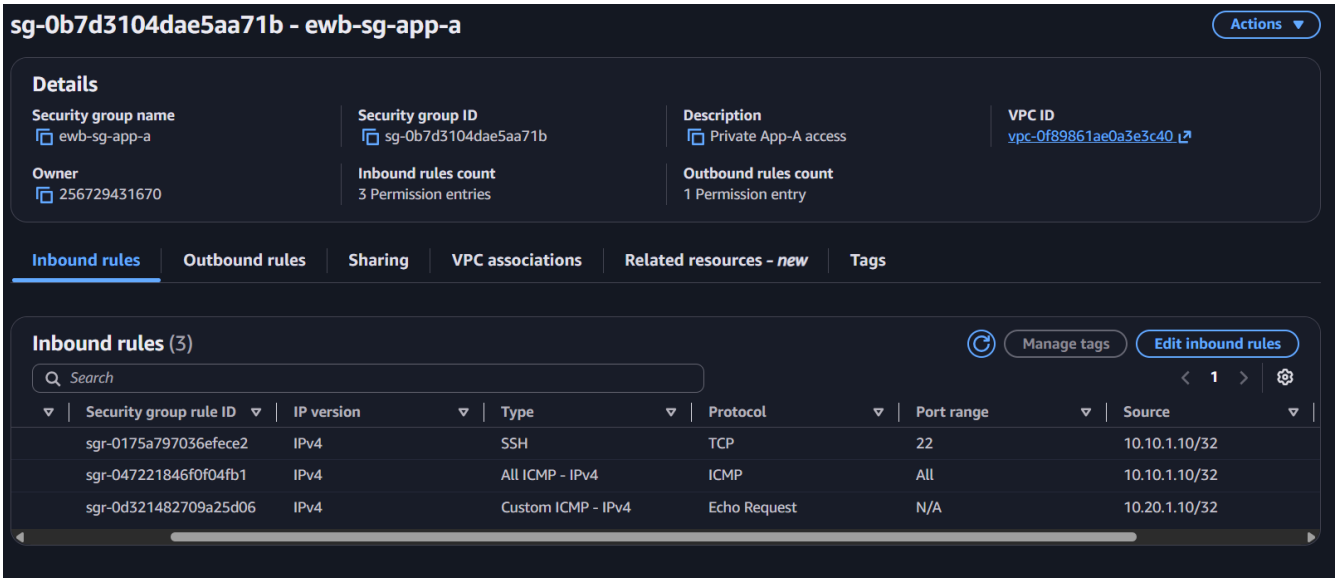


Figure 10. App-B security group

Rules show ICMP Echo Request and SSH restricted to specified private source addresses.

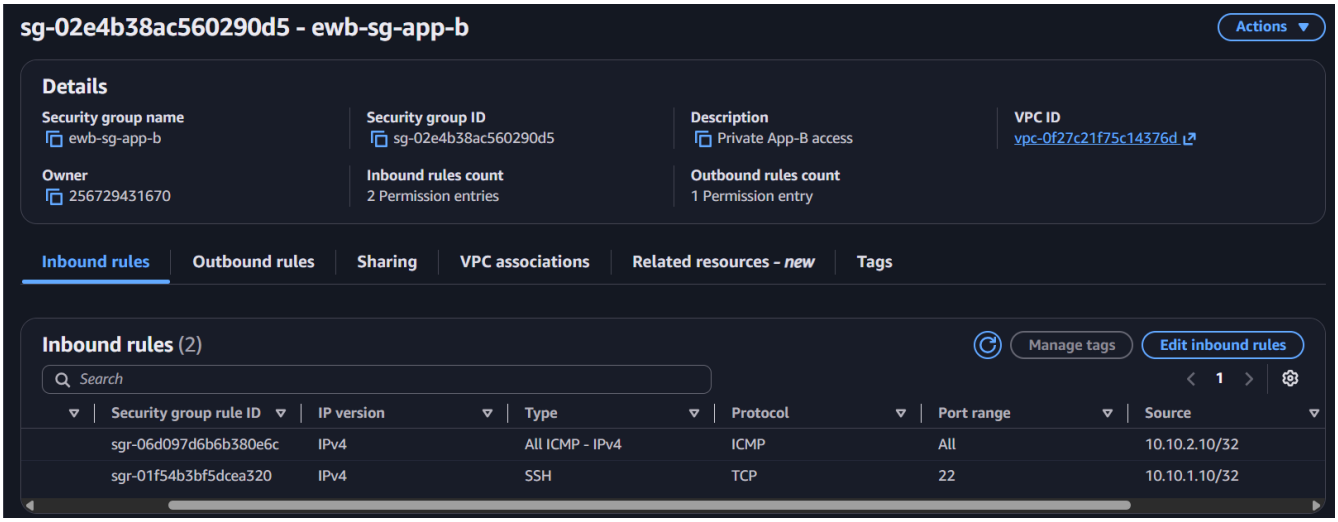


Figure 11. Successful ping tests from the bastion

The terminal shows successful ping replies from App-A (10.10.2.10) and App-B (10.20.1.10).

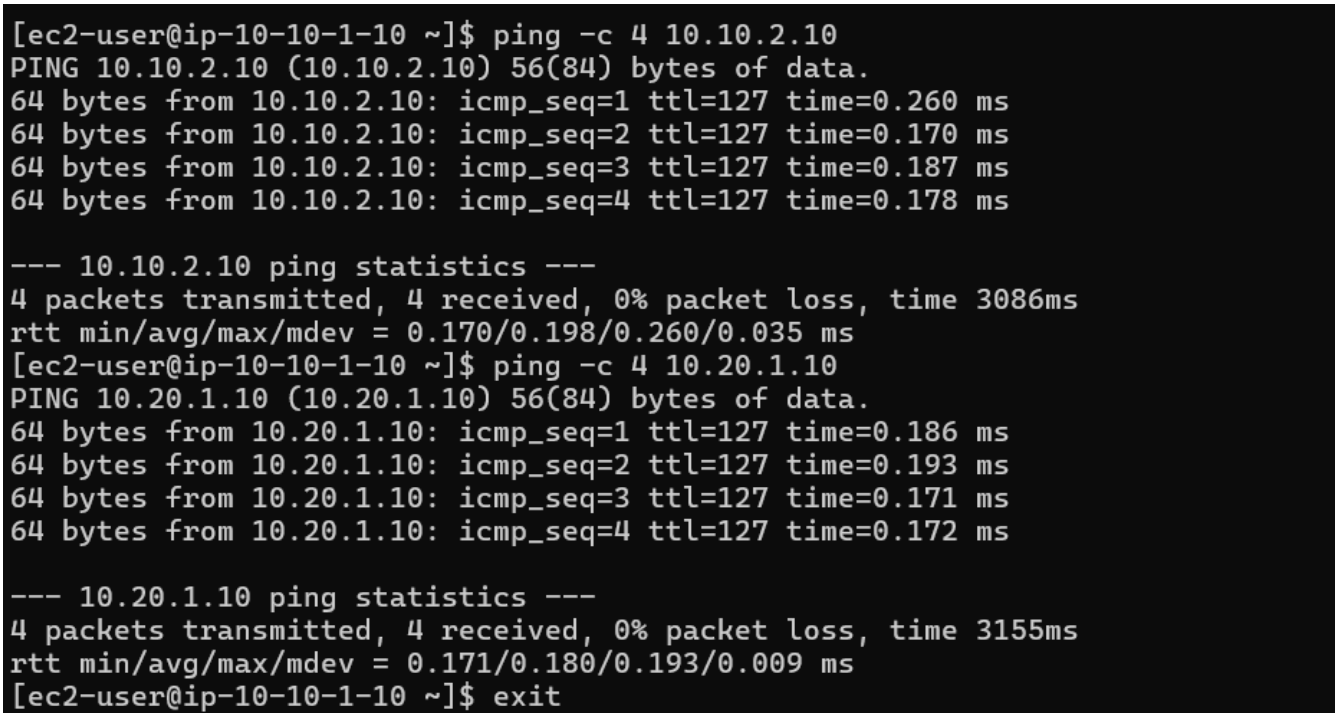


Figure 12. SSH session through the bastion

The terminal evidence shows SSH ProxyCommand use and a private-instance shell prompt.

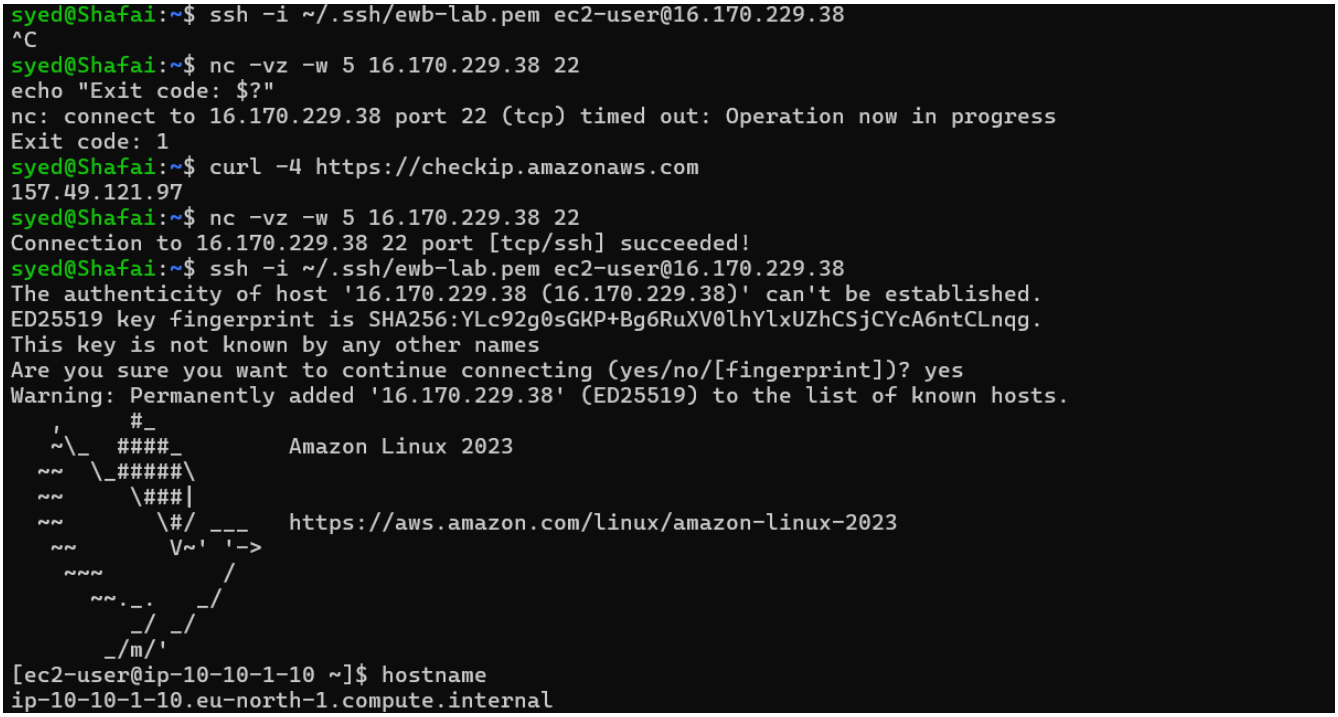


Figure 13. Controlled ICMP failure

The App-A terminal shows 100% packet loss to App-B during the controlled test.

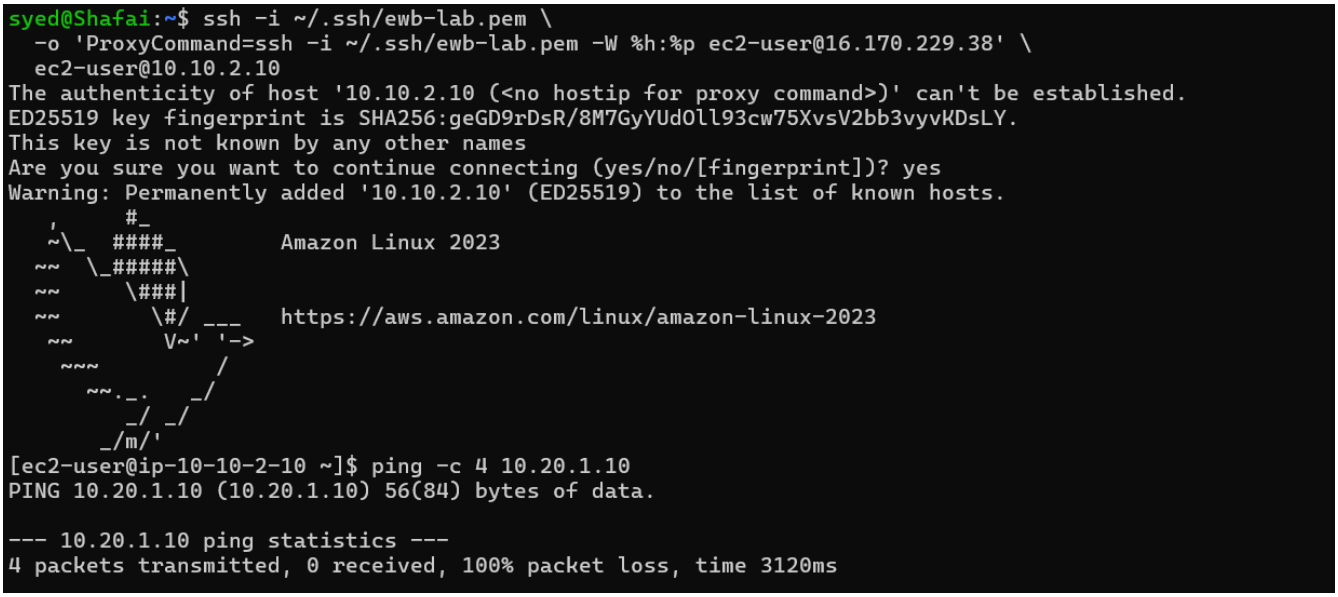
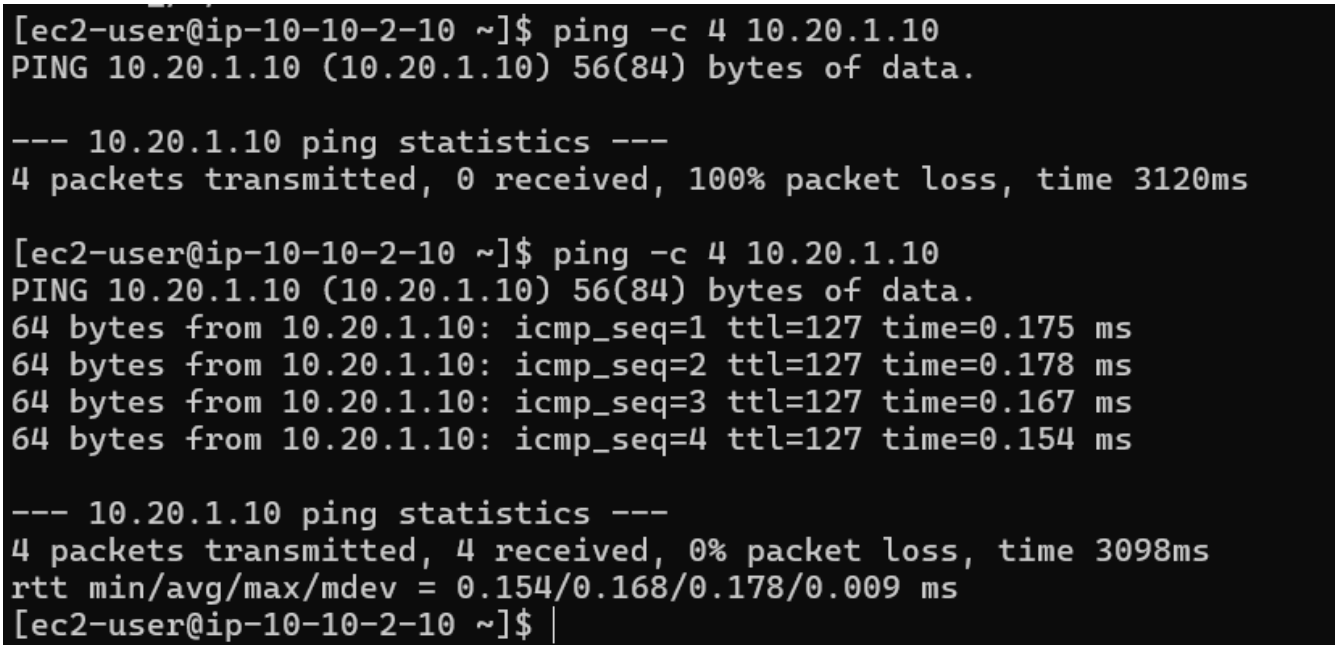


Figure 14. ICMP recovery

The terminal shows failed ping followed by successful ping after the ICMP rule was restored.



4. Conclusion

The evidence demonstrates successful communication between the private instances across the VPC peering connection and successful SSH access to App-B through the bastion. Removing the specific ICMP rule caused the expected ping failure; restoring it returned connectivity. The design keeps App-A and App-B without public IPv4 addresses and uses narrowly scoped inbound rules.

5. Cleanup reminder

Before finishing the lab, save this report and the original screenshots locally. Then follow the lab's cleanup instructions: terminate the three EC2 instances, delete the peering connection, and remove the lab-only route tables, security groups, subnets, internet gateway, and VPCs in dependency order. Check for leftover EBS volumes, Elastic IPs, snapshots, and other chargeable resources. Do not delete resources that belong to other projects.

End of report

Supplemental Evidence & Architecture

Prepared by: Shafai Ahmed Rizvi | AWS Region: Europe (Stockholm), eu-north-1 | 30 October 2026

This supplement adds the detailed route entries, subnet associations, and the latest reverse-ping result to the project report. The architecture below reflects the observed configuration and private IP addressing.

Architecture diagram

VPC A — ewb-vpc-a CIDR: 10.10.0.0/16		**VPC B — ewb-vpc-b** CIDR: 10.20.0.0/16
Public subnet ewb-a-public 10.10.1.0/24 **Bastion** 10.10.1.10 Public IPv4: 16.170.229.38 (observed)	**VPC Peering** ewb-a-b-peer pcx-0370124e0cefd7ad8 Status: Active	**Private subnet** ewb-b-private 10.20.1.0/24 **App-B** 10.20.1.10 No public IPv4
Private subnet ewb-a-private 10.10.2.0/24 **App-A** 10.10.2.10 No public IPv4		**Private routing** VPC B route to 10.10.0.0/16 via peering VPC A routes to 10.20.0.0/16 via peering

Traffic path: App-A traffic destined for 10.20.1.10 uses VPC A's route for 10.20.0.0/16, traverses the active peering connection, and reaches App-B through VPC B's local route. The return traffic requires VPC B's route for 10.10.0.0/16 through the same peering connection. SSH to App-B is tunneled through the public bastion; App-B itself has no public IPv4 address.

Note: The bastion address shown is the observed auto-assigned public IP in the supplied evidence and may change after a stop/start. No Elastic IP is shown.

Detailed route table evidence

All listed routes were shown as Active in the supplied AWS console screenshots.

Route table	Destination	Target
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	10.20.0.0/16	pcx-0370124e0cefd7ad8
ewb-rt-a-public	0.0.0.0/0	igw-05a9b01fc5f873cc5
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	pcx-0370124e0cefd7ad8
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	pcx-0370124e0cefd7ad8

Route table screenshots

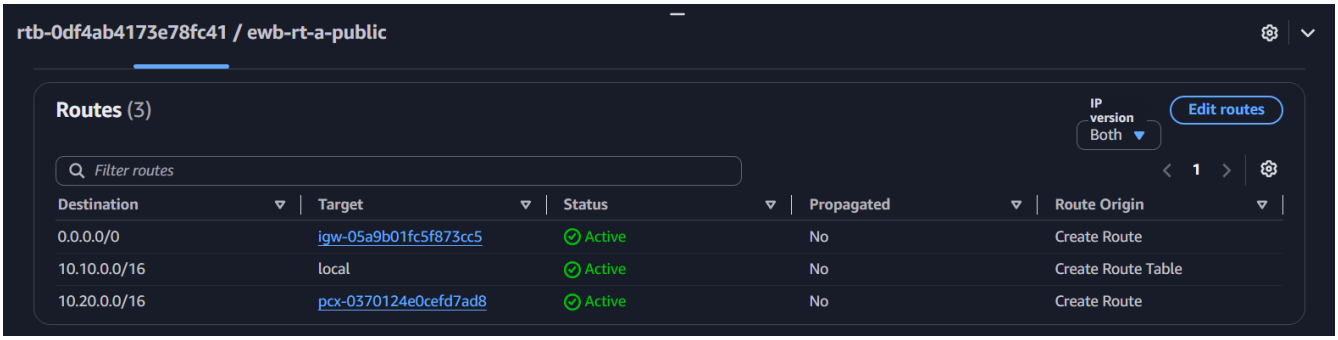


Figure A1. ewb-rt-a-public — local route, peering route, and default route to the Internet Gateway.

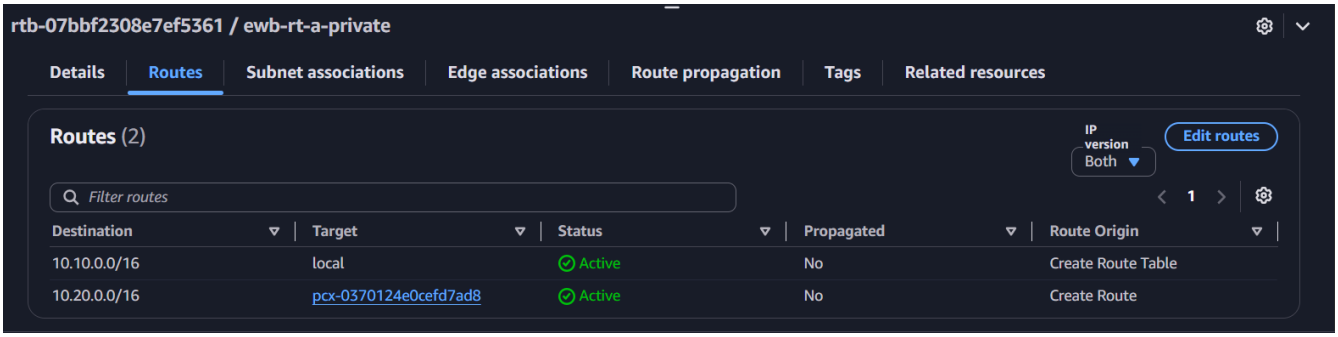


Figure A2. ewb-rt-a-private — local route and peering route to VPC B.

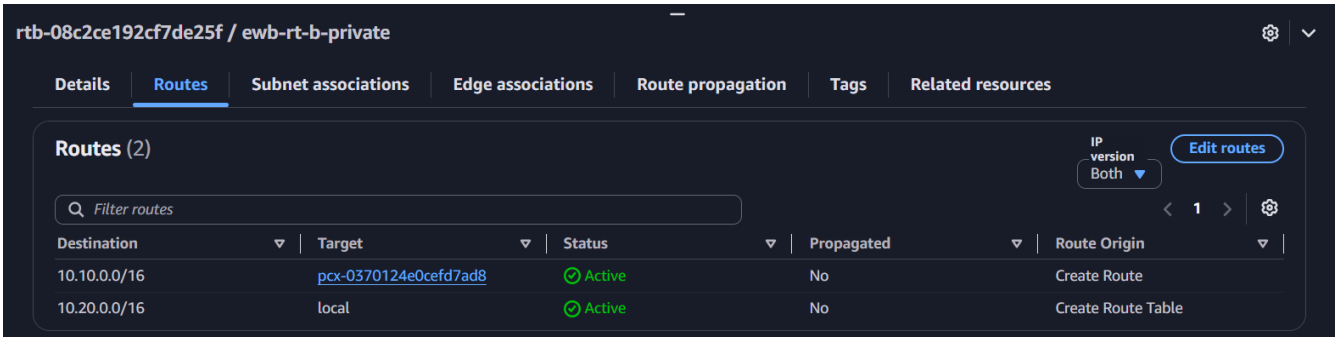


Figure A3. ewb-rt-b-private — local route and peering route to VPC A.

Subnet associations

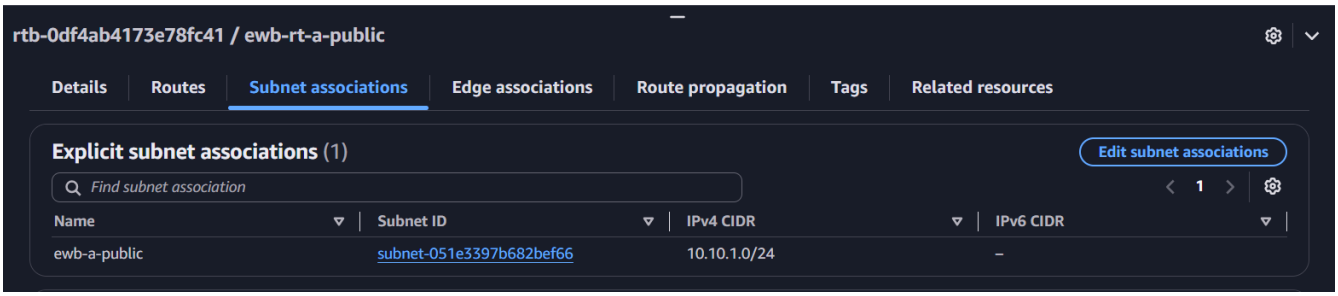


Figure A4. ewb-rt-a-public is associated with ewb-a-public (10.10.1.0/24).

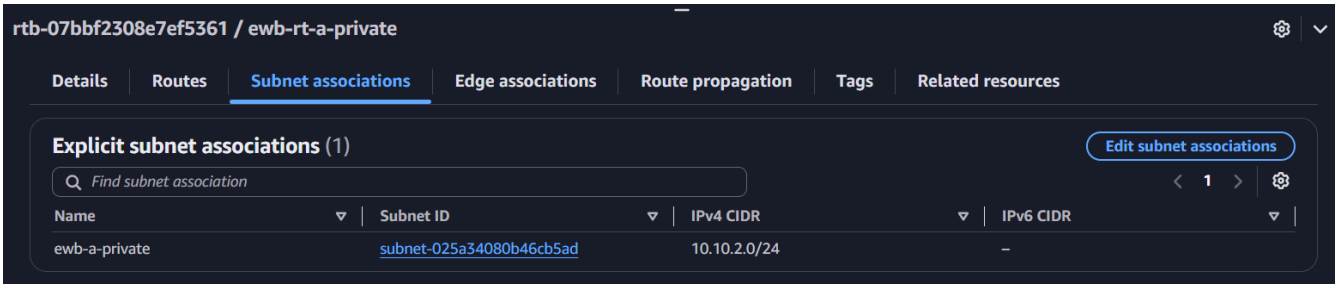


Figure A5. ewb-rt-a-private is associated with ewb-a-private (10.10.2.0/24).

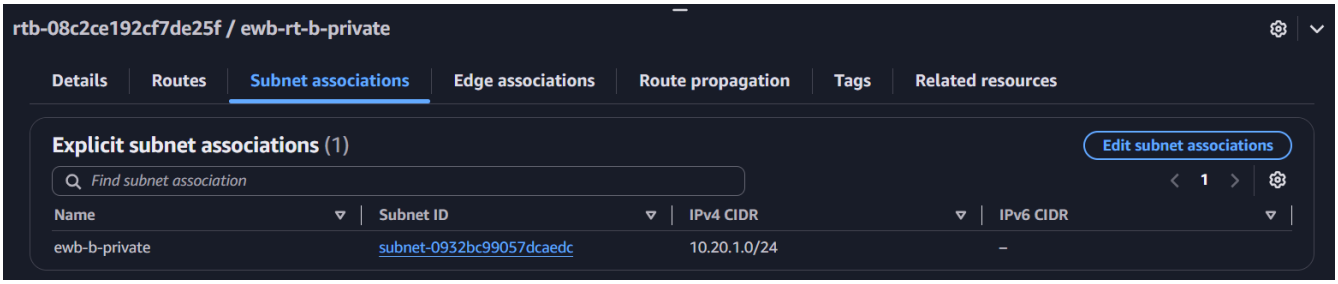


Figure A6. ewb-rt-b-private is associated with ewb-b-private (10.20.1.0/24).

Reverse connectivity test: App-B to App-A

The terminal screenshot shows App-B (10.20.1.10) successfully pinging App-A (10.10.2.10). Four ICMP echo requests received replies; packet loss was 0%. This confirms reverse-direction private connectivity across the peering connection.

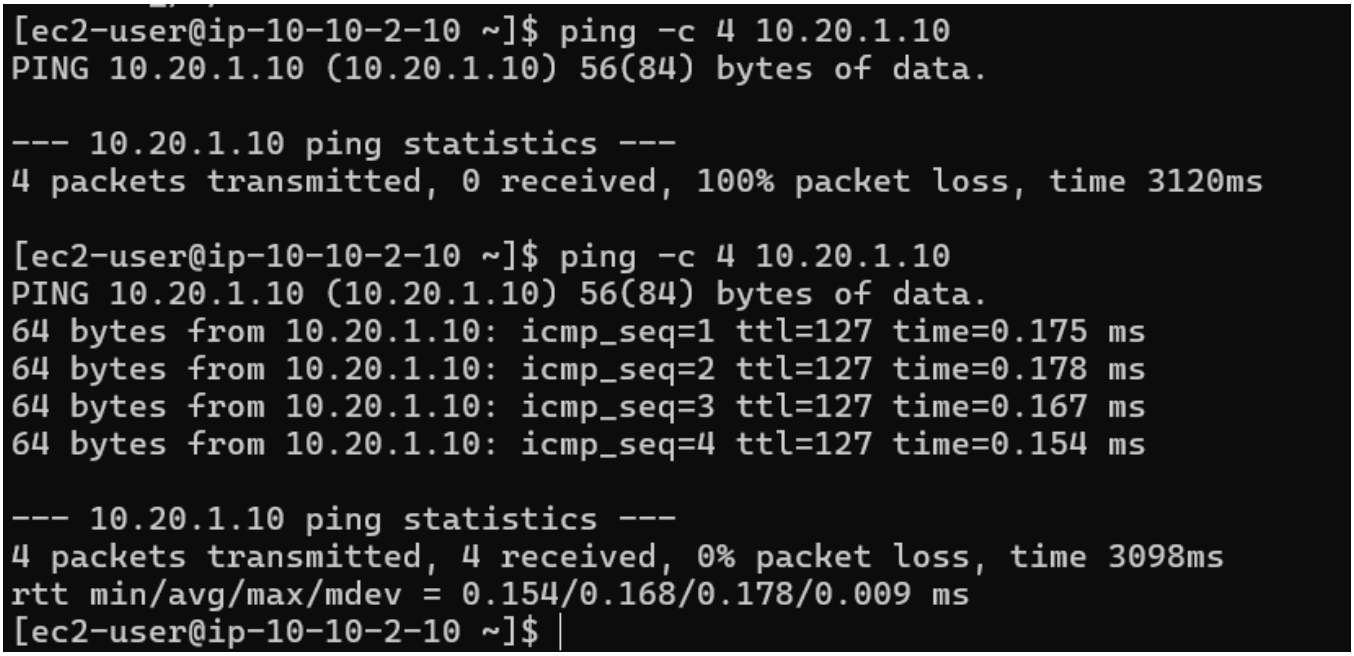


Figure A7. App-B → App-A: 4 packets received, 0% packet loss.

Submission note: The report's earlier sections include the supplied security-group, SSH-through-bastion, and ICMP failure/recovery evidence. The baseline-failure screenshot before peering was not included. Cleanup evidence is not included because resource cleanup has not yet been confirmed; complete cleanup after saving all evidence and, if required by the instructor, add a screenshot of the cleared lab resources.